

Pulleti Sai Sirisha

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PROFESSIONAL SUMMARY

AI/ML professional specializing in architecting, integrating, and operationalizing agentic AI systems for enterprise use cases. Strong expertise in Machine Learning, NLP, model fine-tuning, multi-agent architectures, RAG-based systems, and full-stack AI platforms that integrate LLMs into real-world workflows. Proven experience delivering scalable, production-ready AI solutions across cloud environments, with a focus on automation, intelligent decision-making, and enterprise-grade deployment. Passionate about building reliable and impactful AI systems that drive measurable business outcomes.

EXPERIENCE

Virtusa Consulting Services Private Limited

29th Dec 2023 – Present

Associate Engineer

Hyderabad, Telangana

- Architected and deployed an enterprise-scale Agentic AI platform using Google ADK, Vertex AI Agent Engine, and GCP to proactively identify SLA breach risks across Google Cloud cases, reducing escalations by 70–80% and improving customer satisfaction.
- Implemented a knowledge-grounded RAG pipeline that combined enterprise KB articles with contextual case data to generate root-cause analysis and next-best-action recommendations.
- Integrated Gmail and Google Chat notifications for real-time risk escalation alerts, significantly reducing manual monitoring effort and improving turnaround time.
- Developed a multi-agent SDLC automation platform using LangGraph that converted natural language user stories into production-ready microservices across multiple programming languages, accelerating application development and reducing manual engineering effort.
- Implemented automated test-case generation from requirements, improving test coverage and accelerating software delivery.

PROJECTS

Multimodal Voice & Document Intelligence

Vision API, Speech-to-Text, FastAPI, React.js, MongoDB

- Built an AI-powered pipeline that extracted insights from scanned pitch decks using OCR and from founder interviews using speech-to-text transcription.
- Combined document and voice intelligence to enable faster and more comprehensive startup evaluations.
- Developed a scalable FastAPI, React.js, and MongoDB application for document management, AI workflow orchestration, and real-time report generation, reducing evaluation time from hours to 4–5 minutes.

BERT-Based Sentiment Analysis and AWS Deployment

NLP, Hugging Face, Docker, AWS SageMaker

- Built an end-to-end NLP sentiment analysis system comparing a TF-IDF + Linear SVM baseline with a fine-tuned BERT model, achieving 95% F1-score with BERT and 90% accuracy with SVM.
- Containerized the inference service with Docker and FastAPI, stored artifacts in Amazon S3, pushed the image to Amazon ECR, and deployed a real-time Amazon SageMaker endpoint for low-latency sentiment prediction.

SKILLS

Agentic AI Frameworks: LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, AutoGen, Model Context Protocol (MCP), A2A Protocol

AI / ML & GenAI: Machine Learning, Deep Learning (Neural Networks), NLP, Fine-Tuning, Gemini, GPT Models (Transformer Architecture), RAG Pipelines, Vector databases, Prompt Engineering, LLM Evaluations, Hugging Face

Programming Frameworks & Databases: Python, FastAPI, Flask, React.js, SQL, MongoDB

Cloud Services: GCP Vertex AI, Gemini Enterprise, Guardrails, Cloud Run, AWS Bedrock, Amazon SageMaker, Amazon S3 Bucket, Amazon ECR

DevOps: Docker, CI/CD, Git

CERTIFICATIONS

- Microsoft Certified Azure AI Fundamentals (AI-900)
- AWS Certified AI Practitioner
- Google Cloud Certified Associate Cloud Engineer
- Google Generative AI Leader Certification

ACHIEVEMENTS

- Winner of the Virtusa Agentic AI Engineer Hackathon and awarded Outstanding Performer award for developing innovative multi-agent AI systems leveraging LLM orchestration and scalable AI pipelines to solve complex automation challenges.

EDUCATION

Velagapudi Ramakrishna Siddhartha Engineering College
B-Tech : Electrical and Electronics Engineering (CGPA 9.20)

Vijayawada ,AP
Aug 2019 – July 2023